
Neural Structural Reasoner: A Brain-inspired Architecture for Reasoning over Structured Knowledge

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Abstract

1 Structural reasoning, the ability to recognize and make inference over the relational
2 structure between objects and concepts, is a hallmark of human cognition, yet
3 prevailing methods often collapse relational topology into flat embeddings, cannot
4 discover hidden structure and lack interpretability. We introduce Neural Structural
5 Reasoner (NSR), a brain-inspired network that preserves relational structure di-
6 rectly in the connectivity and dynamics of coupled neuronal populations. NSR
7 draws inspiration from three biological mechanisms: multi-layered architecture for
8 encoding hierarchical knowledge, stable representation of entity and concepts, and
9 path integration for input-driven state inference. At query time, NSR parallelizes
10 computation over candidate relational structures and leverages confidence-weighted
11 scores to perform link prediction. On standard knowledge-graph benchmarks, NSR
12 matches strong embedding and rule-mining baselines with superior training effi-
13 ciency. Because reasoning is implemented through sequences of human-readable
14 neuron activations, NSR affords native interpretability by tracking intermediate
15 inference steps. The model further extracts latent relational hierarchies and compo-
16 sitional rules, demonstrating the brain-inspired architecture as an effective, efficient,
17 and highly interpretable substrate for structural reasoning.

18 1 Introduction

19 Knowledge in the real world is inherently structured—organized by relations such as order, hierarchy,
20 and connectivity—and humans routinely rely on these relations, rather than the entities alone, to
21 reason about the world [18, 20]. This ability to understand and reason about data relationships,
22 which we call *structural reasoning*, has long been regarded as a desideratum for human-like machine
23 intelligence [24, 48]. Machine learning pursues it from several angles: knowledge-graph embeddings
24 map entities and typed relations into continuous spaces for multi-hop queries [22, 23, 47]; graph
25 neural networks impose relational inductive biases via message passing [4, 37, 45]; and large
26 language models, now widely deployed as general reasoners, tackle structural tasks by verbalizing
27 relations and chaining inferences in natural language [52, 21, 16, 31]. Despite their differences, these
28 methods share two recurring weaknesses: they tend to collapse rich structural information into flat
29 vector representations and produce answers without exposing how they were reached—making their
30 reasoning hard to verify and prone to break on deeper, multi-step queries [3, 54].

31 Biology offers an alternative starting point. Structural reasoning in the brain arises from the
32 coordinated activity of several systems: the hippocampal–entorhinal circuit builds cognitive
33 maps—relational organizations of entities that support inference beyond direct experience across
34 spatial and non-spatial domains [5, 50], and does so predictively and in a manner causally re-
35 quired for structure-based inference [40, 28]; the prefrontal cortex abstracts relations into hierarchical,

36 schema-like representations that generalize across contexts [36]. Existing biologically inspired models,
 37 including the Tolman–Eichenbaum Machine [50], Vector-HaSH [10], and related associative-memory
 38 architectures [41, 43, 35], draw on these neural principles to represent and retrieve structured knowl-
 39 edge. However, to our knowledge, there still remains a lack of a modular, brain-inspired architecture
 40 that can both encode and perform reasoning over structured knowledge at scale.

41 We address this gap with Neural Structural Reasoner (NSR), a brain-inspired network architecture for
 42 structural reasoning. The main advances are four-fold:

- 43 • **Network structure as relational structure.** NSR uses network connectivity to represent relational
 44 structure, and exploits efficient network dynamics to implement relational reasoning.
- 45 • **Parallel reasoning with ranked confidence.** NSR computes in parallel over multiple candidate
 46 relational structures, returning ranked candidate answers with confidence score.
- 47 • **Competitive performance with high computational efficiency.** NSR approaches strong ML
 48 baselines on standard structural-reasoning benchmarks while requiring substantially less training
 49 and inference compute.
- 50 • **Interpretability from dynamics.** NSR yields human-readable reasoning paths by tracking network
 51 dynamics as reasoning steps at inference.

52 2 Background and Related Works

53 2.1 Structural reasoning across machine learning

54 Structural reasoning has long been studied under Knowledge Representation and Reasoning
 55 (KRR) [7], which formalizes knowledge through structured objects—logic rules, ontologies, and
 56 graphs—and defines inference as a process operating on that structure. Among these, knowledge
 57 graphs (KGs) offer an instantiation that is both scalable in practice and conducive to systematic
 58 empirical study. Formally, a KG is a tuple $\mathcal{G} = (\mathcal{E}, \mathcal{R}, \mathcal{T})$ with entities $\mathcal{E}$, relations $\mathcal{R}$, and triples
 59 $\mathcal{T} \subseteq \mathcal{E} \times \mathcal{R} \times \mathcal{E}$, where each $(h, r, t) \in \mathcal{T}$ asserts that entity h stands in relation r to entity t
 60 [22, 23]. The canonical task is link prediction: given a query $(h, r, ?)$, score candidate tails and
 61 recover the missing entity from the relational structure already in $\mathcal{G}$. This makes structural reason-
 62 ing concrete—inference over typed relations among discrete entities, isolated from perceptual or
 63 linguistic confounds.

64 A central object of study is the structure of relations themselves, since real KGs are pervaded
 65 by canonical patterns like symmetry (e.g., `sibling_of`), inversion ($r_1(x, y) \Rightarrow r_2(y, x)$, e.g.,
 66 `teacher_of` $\leftrightarrow$ `student_of`), and composition ($r_1(x, y) \wedge r_2(y, z) \Rightarrow r_3(x, z)$, e.g., `father_of` $\circ$
 67 `father_of` $\Rightarrow$ `grandfather_of`) [42]. Composition is especially important: nearly every relation
 68 in benchmark KGs participates in some compositional pattern [26]. Existing methods address
 69 these regularities through distinct representational strategies, each with characteristic limitations.
 70 **Geometric embedding and tensor-factorization models** encode relations as translations, rotations,
 71 bilinear maps, or low-rank interactions, enabling scalable link prediction but leaving composition
 72 implicit in vector geometry [6, 42, 25, 44]. **Deep neural models**, including ConvE [14] and message-
 73 passing GNNs [38, 55, 19], add nonlinear computation and neighborhood aggregation, but often lack
 74 transparent inference trajectories and struggle with long-range relational chains. **Rule-mining and**
 75 **path-based systems** expose logical chains, but rely on combinatorial search and are brittle under
 76 missing or noisy edges [53, 13]. LLM-based methods can verbalize triples or retrieve subgraphs, yet
 77 their outputs are prompt-sensitive and may not faithfully track executable graph operations. Thus,
 78 current methods generally lack a native mechanism for representing relational structure as explicit,
 79 reusable, and traceable dynamics over a structured network.

80 2.2 Neural substrates of structural reasoning

81 Structural reasoning in the brain appears to rely on a distributed network of brain regions. Conceptual
 82 knowledge is often linked to the anterior temporal lobe (ATL), a proposed transmodal semantic hub
 83 that integrates modality-specific features into **stable, context-general concepts** [30, 34]. Relational
 84 operations recruit prefrontal mechanisms, especially rostralateral/frontopolar prefrontal cortex, whose
 85 **activity increases with relational integration** demands and which is implicated in analogical and
 86 multi-relational reasoning [11, 8, 49, 46]. **Flexible composition over relations** further depends

on medial temporal lobe circuitry: the hippocampus binds overlapping experiences into relational structures that support novel inference [15, 32, 17], while entorhinal grid-like codes may provide a metric format for cognitive maps spanning physical and conceptual spaces [12, 29]. Through **path integration**, hippocampal–entorhinal circuits update an internal state estimate by accumulating transitions through space [5]; by extension, analogous dynamics may support abstract reasoning by tracking trajectories through relational or conceptual spaces (though direct evidence remains scarce). Thus, structural reasoning in the brain likely emerges from specialized but interconnected regions that support stable entity and relation representations, relation-specific binding of entities, and compositional generalization of relational structures.

2.3 Biologically inspired models of structural reasoning

Some biologically inspired architectures offer important ingredients for relational cognition, though they have not yet fully addressed structural reasoning in the sense of §2.1. The Tolman–Eichenbaum Machine (TEM) [50, 51] repurposes hippocampal–entorhinal path integration for abstract inference, treating relations as velocity-like signals over a learned structural scaffold; however, its entorhinal module is implemented as a backpropagation-trained RNN, leaving open how such computations arise from grid-cell-like dynamics. Vector–HaSH [10] provides a more biologically grounded HPC–EC loop and supports sequence memory via path integration, but its role in abstract relational reasoning remains unexplored. Continuous-attractor models of grid cells [9] offer a mechanistic account in which velocity-related inputs drive state transition to encode movement in space, yet their application is often limited to low-dimensional navigation rather than high-dimensional relational domains. A deeper limitation of many HPC–EC inspired models is that the network dynamics, built for 2D navigation, are confined to a 2D torus, while **abstract relations require input-driven transitions on high-dimensional manifolds**—and no biologically plausible architecture yet supports this. Together, these works motivate a bio-inspired architectures that can encode and perform reasoning in structured, high-dimensional relation space.

3 The Neural Structural Reasoner Model

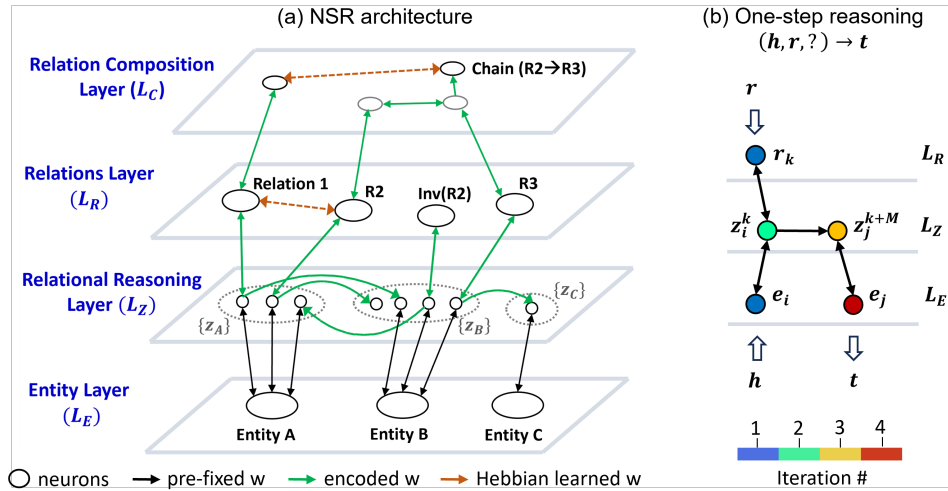


Figure 1: **The Neural Structural Reasoner Architecture.** (a) Four core layers of NSR: L_E encode entities; L_Z instantiates relation-driven state transition; L_R and L_C encode relations and their compositions (e.g., chains). Connections in black are fixed a priori, those in green encode observed triples, those in orange are learned via Hebbian updates to capture latent relational structure. (b) Sequence of neuron activation that implements one-step reasoning over $(h, r, ?) \mapsto t$.

113 3.1 Model Architecture

114 Neural Structural Reasoner (NSR) maps relational structure between entities onto a hierarchical
115 multi-layered network, where connectivity directly encodes associations between entities, relations
116 and relation compositions (Fig. 1a). The core functional layers are:

117 **Entity Layer** (L_E): contains neurons $\{e_i\}^N$ encoding the N entities in the data.

118 **Relations Layer** (L_R): contains a total of $2M$ neurons where the first half, $\{r_i\}^M$, encode the M
119 relations and the second half, $\{r_{i+M}\}^M$, encode the inverse relations $\{r^{-1}\}^M$ (e.g., father_of^{-1}
120 is learned to associate with son_of via the Hebbian rule in §3.3).

121 **Relational Reasoning Layer** (L_Z): contains up to $N \times (2M)$ neurons; a given neuron z_i^k encodes
122 the association between the head entity h_i and the relation r_k .

123 **Relation Composition Layer** (L_C): contains a flexible number of neurons encoding the composi-
124 tional structures (e.g. chains) among relations.

125 3.1.1 Network Connectivity and Dynamics

126 The four layers are connected in a bi-directional hierarchy: $L_E \leftrightarrow L_Z \leftrightarrow L_R \leftrightarrow L_C$. L_E and L_Z
127 neurons encoding the same entity are bidirectionally coupled at initialization and held fixed:

$$\forall i \in [N], \forall k \in [2M] : W_{E \leftrightarrow Z}(e_i, z_i^k) = 1 \quad (1)$$

128 All other weights are initialized at 0 and are learnable. The activation state of each layer is updated in
129 a single step by summing up intra- and inter-layer inputs and external query input I , and transformed
130 through sigmoidal nonlinearity σ .

$$\begin{cases} L_E : X_E(t+1) = \sigma_E(W_{E \leftrightarrow Z}X_Z(t)) + I_E, \\ L_Z : X_Z(t+1) = \sigma_Z(W_{ZZ}H(X_Z(t) - \eta_Z) + W_{Z \leftrightarrow E}X_E(t) + W_{Z \leftrightarrow R}H(X_R(t) - \eta_R)) + I_Z, \\ L_R : X_R(t+1) = \sigma_R(W_{RR}X_R(t) + W_{R \leftrightarrow Z}H(X_Z(t) - \eta_Z) + W_{R \leftrightarrow C}X_C(t)) + I_R, \\ L_C : X_C(t+1) = \sigma_C(W_{CC}X_C(t) + W_{C \leftrightarrow R}H(X_R(t) - \eta_R)) + I_C \end{cases} \quad (2)$$

131 Neuronal interaction within L_Z are rectified by the Heaviside function H , ensuring that activity
132 propagates to only in response to convergent inputs from the associated entity and relation. Similar
133 gating exists for projections $L_Z \rightarrow L_R$ and $L_R \rightarrow L_C$ to conditional activation of co-occurring
134 relations and relation compositions in the learning phase (§3.3). When performing reasoning, the
135 relevant entities and relations obtained in intermediate steps can be read out directly from neuronal
136 activations in L_E , L_R or L_C .

137 3.2 Encoding of Relation Triples

138 For each triple (h_i, r_k, t_j) and its inverse (t_j, r^{-1}, h_i) , we let neurons r_k and r_{k+M} to encode the
139 relation and its inverse, and connect them with the entity-encoding neurons by assigning the following
140 weights :

$$\begin{cases} W_{ZZ}(z_i^k, z_j^{k+M}) = W_{ZZ}(z_j^{k+M}, z_i^k) = 1, \\ W_{Z \leftrightarrow R}(z_i^k, r_k) = 1, \quad W_{Z \leftrightarrow R}(z_j^{k+M}, r_{k+M}) = 1 \end{cases} \quad (3)$$

141 3.2.1 Reasoning over $(h_i, r_k, ?)$ through Path-Integration-Inspired Dynamics

142 When queried with the relation triple $(h_i, r_k, ?)$ (Fig. 1b), NSR first activates the entity neuron e_i
143 and the relation neuron r_k . Iterating over (2) then activates neurons z_i^k and z_j^{k+M} through intra-layer
144 dynamics in L_Z . This in turn activates neuron e_j , which allows the readout of the tail entity t_j . This
145 computational logic is inspired by path integration models proposed for entorhinal grid cells [9, 10].

3.3 Learning Compositional Rules among Relations

Following the encoding phase, NSR uses Hebbian-like associative learning to extract the following compositional structure and equivalence rules among relations:

Relational Equivalence: When queried with two relation triples with the same head entity, e.g. (h_i, r_k, t_j) and (h_i, r_l, t_w) , L_Z neurons encoding the tail entities, z_j^{k+M} and z_w^{l+M} will be activated. If in fact $t_j = t_w$, then the activity of z_j^{k+M} and z_w^{l+M} can be elevated above the threshold η_z through mutual coupling with e_j . Iterating over (2) leads to co-activation of r_{k+M} and r_{l+M} and of r_k and r_l . The connections between co-activated relation neurons are updated via a symmetrized version of Oja’s rule:

$$\Delta w_{ij}^{RR} = \delta_{RR}(r_i * r_j) - \mu_{RR}(r_i^2 + r_j^2)w_{ij}^{RR} \quad (4)$$

Where δ_{RR} and μ_{RR} are learning rates. In practice, all relation triples with the same head entity can be queried at once, and weight update can occur simultaneously for all co-activating r_k s. This ultimately leads to strong coupling between neurons encoding semantically analogous relations.

Symmetric and Inverse Relations: Same network dynamics and learning rule can be applied to extract symmetric and inverse relations. This is achieved by letting $r_l = r_k^{-1}$ in the case of symmetry, and $r_l = r_m^{-1}$ in the case of inverse relations.

Composite relations: We focus on detecting the equivalence between multi-hop relation chains and single triples. Take two-hop chains for example, this amounts to detecting

$$r_p(h_i, x_p) \wedge r_q(x_p, x_q) \Rightarrow r_k^{-1}(h_i, x_q) \leftrightarrow r_k(x_q, h_i), \quad (5)$$

where the bidirectional equivalence follows from Eq. 3: every triple (h, r, t) is encoded together with its inverse (t, r^{-1}, h) . The network discovers such rules by checking whether a two-hop chain returns to the original head entity. Concretely, suppose (h_i, r_p, x_p) is already encoded. NSR computes three inference steps:

Step 1: Activate e_{x_p} (the tail of the known triple) and all relation neurons $\{r_m\}_{m=1}^{2M} \setminus \{r_{p+M}\}$ (all relations except the inverse r_p^{-1}). Iterating dynamics in L_Z activates neurons z_q^m representing candidate tails x_q reachable from x_p via each relation r_m .

Step 2: For each discovered entity x_q , treat it as a new head entity and repeat Step 1, querying $(x_q, r_m, ?)$. This yields a second-hop set of tails $\{x_s\}$ encoded by neurons z_s^m .

Step 3: If any x_s coincides with the original head entity h_i , the network has closed a three-step loop:

$$(h_i, r_p, x_p) \wedge (x_p, r_q, x_q) \wedge (x_q, r_k, h_i). \quad (6)$$

Because the loop returns to h_i , the composition $r_p \circ r_q$ acts as the inverse of r_k , i.e. $r_p \circ r_q \Rightarrow r_k^{-1}$ (Eq. 5). Owing to the bidirectional encoding in Eq. 3, the presence of (x_q, r_k, h_i) is equivalent to the presence of (h_i, r_k^{-1}, x_q) ; hence the discovered chain is registered in L_C as a compositional equivalence between $r_p \rightarrow r_q$ and r_k^{-1} .

Whenever such a closed loop is detected, assign a L_C neuron c_k to represent the relation r_k , and neuron $c_{p \rightarrow q}$ to detect the sequential activation of $r_p \rightarrow r_q$. Update weights according to:

$$\Delta w_{k-pq}^{CC} = \delta_{CC}(c_k * c_{pq}) - \mu_{CC}(c_k^2 + c_{pq}^2)w_{k-pq}^{CC} \quad (7)$$

While these procedures are designed for online sequential learning, for static datasets, the existence of composite structures can be detected statistically and then encoded directly in network weights. We present details of these simplifications along with pseudocodes in Appendix A.2.

3.4 Reasoning Phase

Given a query $(h_i, r_k, ?)$ where $(h_i, r_k, t^*) \notin \mathcal{D}_{\text{train}}$, NSR uncovers t^* by first retrieving equivalent relations r_e or relation compositions C_e to r_k , and then performing relational reasoning over these equivalence sets to obtain candidate tail entities $\tilde{\mathcal{T}} = \{\tilde{t}\}$ with confidence scores:

186 **Step 1 Retrieval of equivalent relations:** This is achieved by first activating neuron r_k encoding the
 187 query relation, and iterating the dynamics of L_R in (2) once to activate neurons $\{r_e\}$ encoding the set
 188 of equivalent relations. A threshold T_thresh gates the propagation, retaining only neurons with
 189 activation at least T_thresh ; their activation levels, which represent how strongly two relations are
 190 associated, serve as confidence scores (i.e., $S_R(r_e) = x_e$).

191 **Step 2 Relational reasoning with $\{r_e\}$:** This is implemented by activating the neuron e_i encoding the
 192 head entity, while holding $\{r_e\}$ active. Iterating over the dynamics of L_Z according to (2) implements
 193 $(h_i, r_e, ?)$ and subsequently activates neurons $\{\tilde{e}\}$ in L_E . These neurons encode a set of candidate
 194 tail entities $\tilde{\mathcal{T}}^{(1)} = \{\tilde{t}\}$.

195 **Step 3 Retrieval of equivalent relation compositions:** Upon re-initializing the network, activate the
 196 query relation neuron r_k and allow its activity to propagate to L_C . Iterating L_C dynamics based on
 197 (2) activates neurons encoding relation compositions that are closely associated with r_k . Applying
 198 a gate function yields the top-K activated neurons $\{c_e\} = \text{TopK}(X_C)$ corresponding to the top
 199 equivalent compositions. The activation of $\{c_e\}$ serves as the confidence scores S_C .

200 **Step 4 Relational reasoning with $\{C_e\}$:** Sample and activate a single neuron in $\{c_e\}$, which then
 201 initiates the sequential activation of L_R neurons $\{r_p, r_{p+1}, \dots, r_{p+q}\}$ in the relation chain. Activate the
 202 entity neuron e_i together with r_p to initiate multi-hop reasoning in L_Z . At the end of the iterations,
 203 extract the last tail entity by identifying the top activating neuron in L_E . This step is repeated for all
 204 neurons in $\{c_e\}$ to collect a second set of candidate tail neurons $\tilde{\mathcal{T}}^{(2)}$.

205 **Step 5 Aggregating confidence scores for candidate entities:** The full set of candidate tail entities
 206 $\tilde{\mathcal{T}} = \tilde{\mathcal{T}}^{(1)} \cup \tilde{\mathcal{T}}^{(2)}$ may include ones that are obtained from multiple reasoning paths. For each
 207 unique entity $t_j \in \tilde{\mathcal{T}}$, let $S_R(t_j)$ and $S_C(t_j)$ denote the highest confidence score it receives from
 208 any equivalent-relation path and any compositional path, respectively. The final score is obtained by
 209 taking the maximum over all reasoning paths:

$$\text{Score}(t_j) = \max\{S_R(t_j), S_C(t_j)\}. \quad (8)$$

210 This max-aggregation ensures that a candidate entity is credited only through its most confident
 211 supporting path, preventing low-confidence alternatives from diluting the prediction. This reasoning
 212 procedure inherently grants **interpretability**, as each activated neuron in L_Z or L_C provides an
 213 explicit trace of the reasoning path.

214 4 Experiments

215 4.1 Experimental setup

216 **Datasets.** We evaluate NSR on four datasets: Countries S3, Nations, and Kinship[24] from the
 217 PyKEEN library [2], alongside our extended benchmark Kinship1990_EXTENDED. Following
 218 NSR’s architecture, we encode entities (e.g., countries in Nations) in the Entity layer (L_E), relations
 219 in the Relation layer (L_R), and entity–relation bindings in the Relational Reasoning layers (L_Z). Kin-
 220 ship1990_EXTENDED is our re-engineered extension of the Kinship dataset [27], expanded to 2240
 221 triples with compositional patterns that require multi-hop reasoning beyond simple inverse relations;
 222 construction details are in Appendix C.2. All datasets are partitioned into train/validation/test splits
 223 under the standard protocol used by ConvE and other KGE baselines [14].

224 **Evaluation.** We adopt the filtered link-prediction setting [6], reporting Mean Reciprocal Rank
 225 (MRR) and Hits@K ($K \in \{1, 3\}$). In addition to predictive accuracy, we measure training time to
 226 assess computational efficiency.

227 4.2 Empirical performance

228 Tables 1 and 2 compare NSR against KG embedding baselines and the neuro-symbolic rule learner
 229 RNNLogic [33]. NSR approaches or exceeds state-of-the-art accuracy on compositional tasks while
 230 training two to four orders of magnitude faster.

231 On Countries S3, test queries ask for a country’s continent, but training triples only link neighbors and
 232 upward locatedIn edges; no country–continent facts are observed directly. NSR scores perfectly

Table 1: Performance comparison on Nations and Countries S3. Best results in each column are highlighted in bold. Hits@ k are reported as percentage values in $[0, 100]$; standard deviations of Nations and Countries S3 are in percentage points ($\times 10^{-2}$).

Model	Nations			Countries S3			Train Time (s)
	MRR $\uparrow$	Hits@1 $\uparrow$	Hits@3 $\uparrow$	MRR $\uparrow$	Hits@1 $\uparrow$	Hits@3 $\uparrow$	Nations, S3
ConvE	0.8029 ± 0.44	69.05 ± 0.64	89.45 ± 0.81	0.1987 ± 2.47	11.25 ± 2.50	19.17 ± 4.04	26.4 ± 1.5 , 20.3 ± 2.7
DistMult	0.6729 ± 1.15	52.94 ± 1.86	75.92 ± 1.78	0.1761 ± 1.42	7.50 ± 1.02	20.42 ± 3.06	15.5 ± 1.3 , 17.4 ± 1.0
RotatE	0.5351 ± 1.46	33.23 ± 2.07	66.02 ± 2.14	0.1184 ± 2.42	4.17 ± 1.86	10.00 ± 4.45	23.6 ± 2.2 , 19.4 ± 2.9
RESCAL	0.4820 ± 3.26	26.17 ± 4.33	60.95 ± 2.65	0.1906 ± 1.00	10.42 ± 2.64	18.33 ± 2.04	19.2 ± 2.9 , 14.7 ± 2.1
ComplEx	0.4174 ± 2.29	20.75 ± 2.66	49.85 ± 3.00	0.0290 ± 2.68	0.83 ± 1.67	2.08 ± 4.17	21.6 ± 1.6 , 18.4 ± 2.2
TransE	0.3924 ± 0.26	3.28 ± 0.51	69.15 ± 1.48	0.1182 ± 0.61	0.00 ± 0.00	14.17 ± 2.76	24.2 ± 2.5 , 18.9 ± 1.7
RNNLogic	0.6168 ± 0.73	41.99 ± 0.40	77.71 ± 2.21	0.3409 ± 3.29	0.00 ± 0.00	62.50 ± 12.36	84.2 ± 4.1 , 5.1 ± 0.0
NSR	0.7938 ± 0.57	66.57 ± 0.86	90.15 ± 0.58	1.0000 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	0.2762 ± 0.0019 , 0.0012 ± 0.0004

Table 2: Performance comparison on Kinship and Kinship1990_EXTENDED. Best results in each column are highlighted in bold. Hits@ k are reported as percentage values in $[0, 100]$; standard deviations of Kinship and Kinship1990_EXTENDED are in percentage points ($\times 10^{-2}$).

Model	Kinship			Kinship1990_EXTENDED			Train Time (s)
	MRR $\uparrow$	Hits@1 $\uparrow$	Hits@3 $\uparrow$	MRR $\uparrow$	Hits@1 $\uparrow$	Hits@3 $\uparrow$	Kinship, Ext.
ConvE	0.7927 ± 0.37	68.11 ± 0.61	88.45 ± 0.25	0.8265 ± 0.91	74.93 ± 1.47	87.83 ± 0.72	62.5 ± 3.1 , 60.5 ± 7.5
DistMult	0.4021 ± 0.63	23.18 ± 0.64	45.63 ± 0.92	0.9409 ± 0.15	90.62 ± 0.30	97.75 ± 0.25	37.3 ± 5.9 , 31.9 ± 4.0
RotatE	0.7598 ± 0.59	63.37 ± 0.94	86.26 ± 0.20	0.5988 ± 4.47	51.94 ± 5.27	64.40 ± 4.20	83.9 ± 4.5 , 26.0 ± 1.1
RESCAL	0.3343 ± 0.98	15.58 ± 1.02	37.57 ± 1.78	0.0144 ± 0.28	0.22 ± 0.12	0.80 ± 0.31	42.0 ± 4.2 , 30.3 ± 5.3
ComplEx	0.3378 ± 0.60	18.62 ± 0.72	37.92 ± 0.69	0.0220 ± 0.34	0.42 ± 0.23	1.41 ± 0.50	40.9 ± 5.6 , 31.4 ± 1.6
TransE	0.2547 ± 0.47	1.14 ± 0.23	38.53 ± 1.19	0.2249 ± 0.58	2.19 ± 0.69	33.42 ± 1.58	62.2 ± 5.7 , 25.3 ± 2.4
RNNLogic	0.7287 ± 0.68	59.68 ± 1.31	83.13 ± 0.70	0.8107 ± 1.82	80.45 ± 1.92	81.38 ± 1.78	61.2 ± 6.5 , 9.3 ± 1.4
NSR	0.6011 ± 0.60	44.17 ± 0.98	70.00 ± 0.74	0.9533 ± 0.30	94.33 ± 0.48	96.47 ± 0.09	0.3850 ± 0.0223 , 0.0451 ± 0.0002

(MRR 1.000) because its dynamics allows sequentially traversal of intermediate entities—neighbor, region, continent—via the learned relational chain. Embedding models fail because static vector similarity cannot recover unobserved multi-step paths (ConvE 0.199, ComplEx 0.029); RNNLogic reaches only 0.341 because externally mined templates miss the compositional regularity that NSR captures through local Hebbian updates.

The most telling contrast is Kinship versus its extended variant. On classical Kinship, ConvE leads (0.793) and RNNLogic scores 0.729, while NSR manages 0.601. On Kinship1990_EXTENDED, however, where novel compositional relations such as *cousin_chain* require reusing primitives in new ways, ConvE drops to 0.827 and RNNLogic to 0.811, while NSR rises to 0.953—surpassing all baselines. This inversion shows that embedding and externally induced rules adapt poorly to novel relational grammar, whereas NSR’s locally learned compositional rules generalize without retraining.

Across all benchmarks, NSR trains in under 0.4 s (from 0.0012 s on Countries S3 to 0.385 s on Kinship) versus 5.1–84.2 s for RNNLogic and 15.5–83.9 s for embedding baselines. This gain in training efficiency stems from replacing iterative backpropagation with local Hebbian updates.

4.3 Traceable reasoning steps

Unlike embedding baselines that rely on post-hoc attention or gradient saliency to explain opaque scalar scores, NSR’s interpretability is intrinsic: the activation trajectory *is* the computation. For the query (Brazil, embassy, ?) (Figure 2), Step 1 of §3.4 clamps embassy onto the L_R layer and iterates the L_R dynamics once through W_{RR} . The Hebbian couplings learned during training (Eq. (4)) activate $\text{weightedunvote}^{-1}$ as the dominant equivalent relation, and its activation level serves as the confidence score S_R . Step 2 then activates the head entity Brazil in L_E while holding $\text{weightedunvote}^{-1}$ active in L_R ; the joint input projects onto the L_Z relation map via the fixed couplings $W_{E \leftrightarrow Z}$ and $W_{Z \leftrightarrow R}$, forming the initial state $\mathbf{X}_Z^{\text{init}}$. Recurrent dynamics through W_{ZZ} concentrate activation onto the neuron $z_{\text{UK}}^{\text{weightedunvote}^{-1}}$, which is read out in L_E as the tail entity UK.

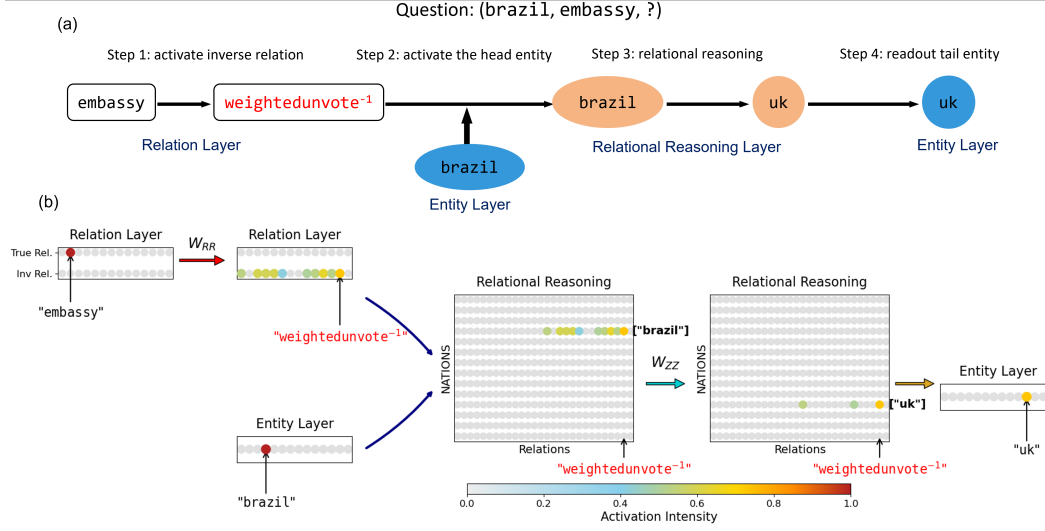


Figure 2: **Traceable reasoning in NSR.** (a) A schematic of the inferred logical path: embassy triggers its inverse to map Brazil to UK. (b) Neuronal activations at each stage of the query (Brazil, embassy, ?). Activating the relation neuron embassy triggers the activation of candidate equivalent relations (e.g., $\text{weightedunvote}^{-1} = \text{embassy}$) in the L_R layer. The selected relation and head entity Brazil jointly project onto the L_Z layer. State transition in in L_Z through recurrent dynamics leads to activation of neurons encoding to the tail entity UK.

Because each active unit represents a discrete, human-readable proposition, errors are auditable by inspecting which relation was activated in L_R or where the L_Z dynamics diverged. This mirrors the replay of structured trajectories in the hippocampal–entorhinal circuit [39], making traceability a biological consequence of the architecture rather than an engineering add-on. Symmetric trace is given in Appendix D.1.

4.4 Learning of latent structures

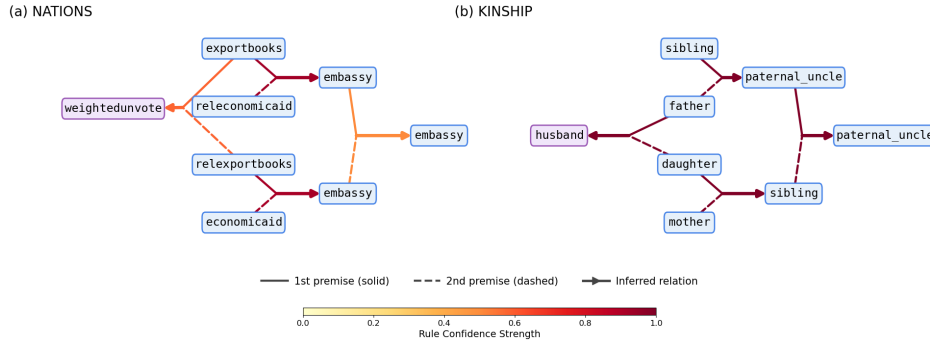


Figure 3: **Latent relational hierarchy.** Compositional rules learned by NSR, satisfying $(h, r_1, x) \wedge (x, r_2, t) = (h, r_3, t)$, on (a) Nations and (b) Kinship1990_EXTENDED. Solid lines denote r_1 , dashed lines denote r_2 , and color intensity indicates rule confidence.

Latent relational hierarchy. NSR recovers the latent compositional structure of the relational space rather than memorizing pairwise co-occurrences. As shown in Figure 3, the model learns which relation sequences form reliable multi-hop pathways. For example, the chain $\text{exportbooks} + \text{releconomicaid} \rightarrow \text{embassy}$ attains confidence 0.93 because the intermediate transition constitutes a coherent diplomatic pathway, whereas merely repeating embassy twice fails to yield a high-

confidence rule despite the relation’s high frequency. This selectivity shows that composition is context-sensitive and depends on whether a relation pair creates a semantically stable bridge. Notably, on Kinship1990_EXTENDED the model can assign confidence 1 to deterministic kinship rules after a single observation by integrating prior logical constraints through hyperparameters, demonstrating that NSR is not limited to statistical induction.

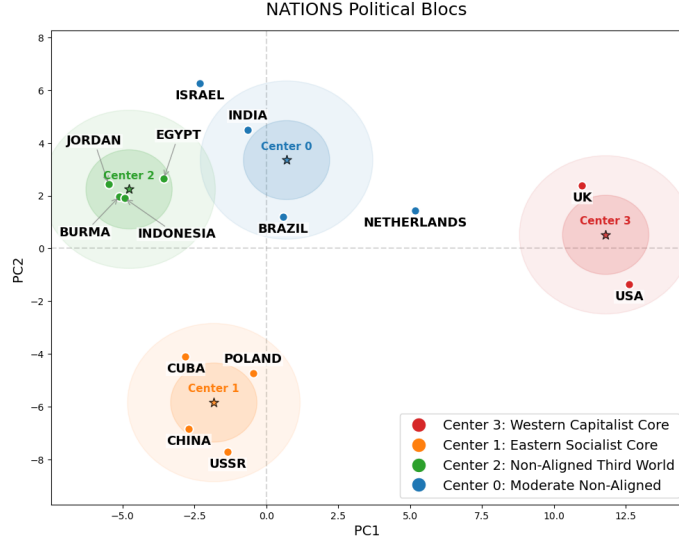


Figure 4: **Learned relational organization in Nations.** PCA projection of nations based on their relational connections in L_Z . Four clusters emerge corresponding to Cold War political blocs.

Relational organization of entities. NSR encodes entities not as isolated symbols but as nodes characterized by their recurrent connections L_Z . As illustrated in Figure 4, projecting entities based on their relational connections (W_{ZZ}) induces structurally meaningful clusters: the Western capitalist core (UK, USA), the Eastern socialist core (USSR, China), and distinct non-aligned blocs (Jordan, Egypt, India, Brazil) each occupy separate regions of the latent space. Taken together, NSR learns a dual latent structure: a compositional hierarchy governing how relations combine, and a relational topology governing how entities are organized.

5 Conclusion

We presented Neural Structural Reasoner (NSR), a brain-inspired architecture that integrates hierarchical representation with relation-driven state transition and Hebbian-type learning to accomplish structural reasoning over knowledge graphs. By preserving relational structure in layered network connectivity rather than collapsing it into flat embeddings, NSR achieves competitive accuracy on standard benchmarks and superior training efficiency, while its inference dynamics natively expose human-readable reasoning traces. NSR also spontaneously discovers interpretable latent structures such as compositional rules among relations and relational similarity among entities. These properties make NSR a promising step towards building efficient and interpretable models for structural reasoning.

NSR is currently evaluated on discrete triplet-structured knowledge graphs, which provide a controlled setting for structural reasoning but leave open whether the same mechanisms generalize to temporal, hyper-relational, event-based, or noisy real-world knowledge. The model also assumes that entities and relations are already symbolically specified, so future work should couple NSR with neural modules that extract relational structure from text, perception, or episodic experience. While NSR is efficient on current benchmarks, scaling compositional-rule discovery to larger graphs and longer multi-hop chains will require sparse activation, approximate retrieval, or learned proposal mechanisms. Future work should also test robustness to missing or contradictory triples and develop more biologically grounded implementations of sequence detection, gating, and confidence aggregation.

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A Details for Encoding and Learning

A.1 Learning and Encoding of KG Training Data

This subsection details how NSR stores the training graph into its connectivity and extracts pairwise relational structure. Compositional multi-hop rules are deferred to §A.2.

Network initialization. All plastic weights are initialized to zero. The only fixed connections are the bidirectional couplings between each entity neuron e_i and its associated L_Z neurons z_i^k (Eq. (1)).

Encoding and pairwise learning. Each training triple (h_i, r_k, t_j) and its inverse (t_j, r_k^{-1}, h_i) are encoded by setting the corresponding L_Z intra-layer and L_Z-L_R weights to one (Eq. (3)). Concurrently, the network records which relations co-occur on the same (h, t) pair; these co-activations drive the Oja-like updates of W_{RR} (Eq. (4)) that capture equivalence, symmetry, and inverse structure.

Algorithm 1 Learning and encoding of KG training data

Input. Training triples $\mathcal{T}_{\text{train}}$; N entities; M relations.

Parameters. Hebbian rate δ_{RR} , decay μ_{RR} .

```

1: initialize all weight matrices to 0
2: for  $i = 1, \dots, N$  do
3:   for  $k = 1, \dots, 2M$  do
4:     fix  $W_{E \leftrightarrow Z}(e_i, z_i^k) = 1$ 
5:   end for
6: end for
7: for each  $(h_i, r_k, t_j) \in \mathcal{T}_{\text{train}}$  do
8:   set  $W_{ZZ}(z_i^k, z_j^{k+M}) = W_{ZZ}(z_j^{k+M}, z_i^k) = 1$ 
9:   set  $W_{Z \leftrightarrow R}(z_i^k, r_k) = W_{Z \leftrightarrow R}(z_j^{k+M}, r_{k+M}) = 1$ 
10:  record  $r_k$  for pair  $(h_i, t_j)$ 
11: end for
12: for each entity pair  $(h, t)$  with  $\geq 2$  recorded relations do
13:   for each distinct pair  $(r_i, r_j)$  in that set do
14:      $\Delta w_{ij}^{RR} \leftarrow \delta_{RR}(r_i r_j) - \mu_{RR}(r_i^2 + r_j^2) w_{ij}^{RR}$ 
15:      $w_{ij}^{RR} \leftarrow w_{ij}^{RR} + \Delta w_{ij}^{RR}$ 
16:   end for
17: end for

```

A.2 Compositional Rules Learning

The Relation Composition layer L_C contains sequence-selective neurons $\{c_{ij}\}$, each tuned to an ordered pair of extended relations $(\tilde{r}_i, \tilde{r}_j)$ with $\tilde{r} \in \{0, \dots, 2M-1\}$ (forward $\tilde{r}_k \equiv r_k$ for $k < M$, inverse $\tilde{r}_{k+M} \equiv r_k^{-1}$). These neurons detect relational chains through asymmetric-delay coincidence detection.

Detection and encoding of relation composition. When the conditions in 5 is satisfied, assign L_C neurons c_p , c_q and c_k to encode relations r_p , r_q and r_k respectively. In addition, assign neuron $c_{\tilde{p}q}$ to be directly coupled with c_q and indirectly coupled with c_p through an intermediate neuron c_s . The activation threshold of $c_{\tilde{p}q}$ through the sigmoidal nonlinearity renders it responsive only when c_p and c_q are activated in a sequential manner. This enables $c_{\tilde{p}q}$ to serve as a sequence detection neuron in a way that is functionally analogous to direction-selective motion detection models proposed for the visual cortex.

Hebbian learning of compositional rules. When a sequence neuron c_{pq} fires upon detecting the ordered pair (r_p, r_q) , and the target relation neuron r_k is concurrently activated (either by a direct triple or by its inverse through the bidirectional encoding of Eq. (3)), the coincident activity strengthens the synapse between c_{pq} and r_k via the same Oja-like rule used in L_R :

$$\Delta w_{k-pq}^{CC} = \delta_{CC} c_k c_{pq} - \mu_{CC} (c_k^2 + c_{pq}^2) w_{k-pq}^{CC}. \quad (9)$$

where the decay term effectively controls for the baseline occurrence rate of relations and chains.

515 The composition-detection and Hebbian mechanisms described above operate locally on individual
 516 triple chains. To discover all compositional rules supported by the data, NSR enumerates every
 517 two-hop path in $\mathcal{T}$ and applies the closed-loop check of Eq. (5) together with the weight update of
 518 Eq. (7). Algorithm 2 implements this global traversal.

Algorithm 2 Compositional-rule learning

Notation. $\mathcal{T}$: training triples (h, r, t) with $r \in \{0, \dots, M-1\}$; extended relations $\tilde{r} \in \{0, \dots, 2M-1\}$ encode forward ($\tilde{r}_k \equiv r_k$) and inverse ($\tilde{r}_{k+M} \equiv r_k^{-1}$) links via Eq. (3). c_{pq} : sequence neuron for (r_p, r_q) . c_k : neuron in L_C representing relation r_k . δ_{CC}, μ_{CC} : Hebbian rate and decay for L_C synapses.

```

1: for each  $(h_i, r_p, x_p) \in \mathcal{T}$  do
2:   for each  $(x_p, r_q, x_q) \in \mathcal{T}$  with  $r_q \neq r_p^{-1}$  do
3:     for each  $(x_q, r_k, h_i) \in \mathcal{T}$  do
4:        $c_{pq} \leftarrow$  sequence neuron for  $(r_p, r_q)$ 
5:        $c_k \leftarrow$  relation neuron for  $r_k$ 
6:        $\Delta w_{k-pq}^{CC} \leftarrow \delta_{CC} c_k c_{pq} - \mu_{CC} (c_k^2 + c_{pq}^2) w_{k-pq}^{CC}$ 
7:        $w_{k-pq}^{CC} \leftarrow w_{k-pq}^{CC} + \Delta w_{k-pq}^{CC}$ 
8:     end for
9:   end for
10: end for

```

519 **A.2.1 Pseudocode for Simplified Learning of Relation Compositions**

520 This version of the algorithm builds on the same network structure and connectivity as Algorithm 2,
 521 but simplifies the relational composition detection and Hebbian learning into faster statistical opera-
 522 tions. The correspondence is as follows:

- 523 (i) **Composition detection.** The condition for detecting the compositional rule $(h, r_i, x) \wedge$
 524 $(x, r_j, t) \leftrightarrow (h, r_k, t)$ is performed by enumerating two-hop chains linking h and t .
- 525 (ii) **Hebbian learning of W_{CC} .** The incremental Hebbian update (line 6 of Algorithm 2) is
 526 emulated by co-occurrence statistics above a support threshold: a rule is written to $\mathcal{C}$ only if
 527 its support exceeds τ_s and its confidence exceeds τ_c . This statistical thresholding mirrors the
 528 effect of the quadratic decay term μ_{CC} in the online dynamics, which drives low-frequency
 529 co-activations to near-zero weights.

Algorithm 3 Simplified compositional-rule learning

Notation. $\mathcal{P}_k = \{(h, t) \mid (h, r_k, t) \in \mathcal{T}_{\text{train}}\}$: ordered entity pairs directly linked by r_k . $\mathcal{S}$ for (r_i, r_j) : all (h, t) reachable via an intermediate x with $(h, x) \in \mathcal{P}_i$ and $(x, t) \in \mathcal{P}_j$. r^* : the relation maximizing $|\mathcal{S} \cap \mathcal{P}_k|$. $\text{conf} = |\mathcal{S} \cap \mathcal{P}_{r^*}|/|\mathcal{S}|$: fraction of chain pairs explained by r^* . **Hyperparameters.** τ_s (*support threshold*): minimum $|\mathcal{S}|$ to consider a chain. τ_c (*confidence threshold*): minimum conf to retain a rule.

```

1:  $\mathcal{C} \leftarrow \emptyset$ 
2: for  $k = 1, \dots, M$  do
3:    $\mathcal{P}_k \leftarrow \{(h, t) \mid (h, r_k, t) \in \mathcal{T}_{\text{train}}\}$ 
4: end for
5: for each pair  $(r_i, r_j)$ ,  $i, j \in \{1, \dots, M\}$  do
6:    $\mathcal{S} \leftarrow \{(h, t) \mid \exists x : (h, x) \in \mathcal{P}_i \text{ and } (x, t) \in \mathcal{P}_j\}$ 
7:   if  $|\mathcal{S}| < \tau_s$  then continue
8:    $r^* \leftarrow \arg \max_{r_k} |\mathcal{S} \cap \mathcal{P}_k|$ 
9:    $\text{conf} \leftarrow |\mathcal{S} \cap \mathcal{P}_{r^*}|/|\mathcal{S}|$ 
10:  if  $\text{conf} \geq \tau_c$  then  $\mathcal{C} \leftarrow \mathcal{C} \cup \{(r_i \circ r_j \Rightarrow r^*, \text{conf})\}$ 
11:   $w_{k-ij}^{CC} \leftarrow \mathcal{C}$ 
12: end for
13: return  $\mathcal{C}$ 

```

530 **B Details for the Reasoning Phase**

531 **B.1 Reasoning phase (algorithmic form)**

532 Algorithm 4 restates the five-step procedure of §3.4 in pseudocode. Layer updates follow the dynamics
 533 defined in Eq. (2); the final aggregation follows Eq. (8).

Algorithm 4 Reasoning phase for query $(h_i, r_k, ?)$.

Require: Query head h_i , query relation r_k ; network weights $W_{E \leftrightarrow Z}, W_{ZZ}, W_{Z \leftrightarrow R}, W_{RR}, W_{R \leftrightarrow C}, W_{CC}$; nonlinearities $\sigma_E, \sigma_Z, \sigma_R, \sigma_C$ and thresholds η_Z, η_R ; activation threshold T_thresh ; gate function $G = \text{TopK}$.

Ensure: Scored candidate tails $\text{Score}(t)$.

```

1: {Step 1: Retrieve equivalent relations from  $L_R$ }
2: Initialize all layer activations  $X_E, X_Z, X_R, X_C \leftarrow \mathbf{0}$ 
3: Clamp query relation:  $X_R[r_k] \leftarrow 1$ 
4: Update  $L_R$ :
5:    $X_R \leftarrow \sigma_R(W_{RR}X_R + W_{Z \leftrightarrow R}H(X_Z - \eta_Z) + W_{R \leftrightarrow C}X_C) + I_R$ 
6:  $\mathcal{R}_{eq} \leftarrow \{r_e \mid X_R[r_e] \geq T\_thresh\}$ 
7:  $S_R(r_e) \leftarrow X_R[r_e]$  for each  $r_e \in \mathcal{R}_{eq}$ 
8: {Step 2: Relational reasoning with  $\mathcal{R}_{eq}$  via  $L_Z$ }
9: Reset  $X_E, X_Z \leftarrow \mathbf{0}$ ; keep  $X_R$  from Step 1
10: Clamp head entity:  $X_E[e_i] \leftarrow 1$ 
11: Update  $L_Z$ :
12:    $X_Z \leftarrow \sigma_Z(W_{ZZ}H(X_Z - \eta_Z) + W_{E \leftrightarrow Z}X_E + W_{R \leftrightarrow Z}X_R) + I_Z$ 
13: Read out  $L_E$ :  $X_E \leftarrow \sigma_E(W_{E \leftrightarrow Z}X_Z) + I_E$ 
14:  $\tilde{T}^{(1)} \leftarrow \{t \mid X_E[t] > 0\}$ 
15:  $S_R(t) \leftarrow X_E[t]$  for each  $t \in \tilde{T}^{(1)}$ 
16: {Step 3: Retrieve relation compositions from  $L_C$ }
17: Re-initialize  $X_E, X_Z, X_R, X_C \leftarrow \mathbf{0}$ 
18: Clamp query relation:  $X_R[r_k] \leftarrow 1$ 
19: Update  $L_C$ :
20:    $X_C \leftarrow \sigma_C(W_{CC}X_C + W_{C \leftrightarrow R}H(X_R - \eta_R)) + I_C$ 
21:  $\{c_e\} \leftarrow G(X_C) = \text{TopK}(X_C)$ 
22:  $S_C(c_e) \leftarrow X_C[c_e]$  for each  $c_e \in \{c_e\}$ 
23: {Step 4: Multi-hop reasoning with each composition neuron}
24:  $\tilde{T}^{(2)} \leftarrow \emptyset$ 
25: for each  $c_e \in \{c_e\}$  do
26:   Reset  $X_E, X_Z, X_R \leftarrow \mathbf{0}$ 
27:   Decode  $c_e$  into sequential relation chain  $\{r_p, r_{p+1}, \dots, r_{p+q}\}$ 
28:   Clamp  $e_i$  in  $L_E$  and  $r_p$  in  $L_R$ 
29:   for each relation index  $m = p$  to  $p + q$  in the chain do
30:     Update  $L_Z$  and read out  $L_E$ 
31:     Advance to next relation  $r_{m+1}$  in  $L_R$  via back-projections  $W_{C \leftrightarrow R}$ 
32:   end for
33:   Extract tail  $t$  from the top-activating neuron in  $L_E$ 
34:    $\tilde{T}^{(2)} \leftarrow \tilde{T}^{(2)} \cup \{t\}$ 
35:   Record  $S_C(t) \leftarrow S_C(c_e)$ 
36: end for
37: {Step 5: Max-aggregation of confidence scores}
38:  $\tilde{T} \leftarrow \tilde{T}^{(1)} \cup \tilde{T}^{(2)}$ 
39: for each unique entity  $t \in \tilde{T}$  do
40:    $S_R(t) \leftarrow \max \{ \text{scores of } t \text{ from Step 2} \}$ 
41:    $S_C(t) \leftarrow \max \{ \text{scores of } t \text{ from Step 4} \}$ 
42:    $\text{Score}(t) \leftarrow \max \{ S_R(t), S_C(t) \}$ 
43: end for
44: return Score

```

534 B.2 Hyperparameter settings

535 Table 3 lists the hyperparameters of the simplified variant used in §3. They are grouped by the
536 processing stage to which they belong.

537 **Reasoning threshold.** T_thresh is the activation threshold of Step 1 (§3.4): a relation neuron
538 r_e enters the equivalent-relation set $\mathcal{R}_{eq}$ only if its activation satisfies $x_e \geq T_thresh$. Given that
539 the Countries dataset contains only two relations, reasoning does not depend on relation similarity;
540 hence, T_thresh is set to infinity.

541 **Learning rate.** The simplified implementation collapses the Hebbian rates δ_{RR} (Eq. (4)) and δ_{CC}
542 (Eq. (7)) to a single scalar δ_θ , because the simplified variant performs one-pass counting instead of
543 iterative Oja updates. The value reported in Table 3 therefore serves as the effective learning rate for
544 both W_{RR} and W_{CC} .

545 **Compositional-rule thresholds** τ_s is the support threshold and τ_c is the confidence threshold τ_c of
546 Algorithm 3: a compositional rule ($r_i \circ r_j \Rightarrow r^*$) is retained only if its entity-pair support $|\mathcal{S}|$ exceeds
547 τ_s and its confidence $\text{conf} = |\mathcal{S} \cap \mathcal{P}_{r^*}|/|\mathcal{S}|$ exceeds τ_c . All values were obtained with Optuna [1];
548 the search ranges were $T_thresh \in [0.05, 0.5]$, $\delta_\theta \in [0.1, 1.0]$, $\tau_s \in [1, 50]$, and $\tau_c \in [0.05, 0.5]$.
549 NSR is not highly sensitive to these settings.

Table 3: Hyperparameters for the main KG benchmarks

Dataset	T_thresh	τ_s	τ_c	δ_θ
Nations	0.22	23	0.20	0.52
Kinship	0.90	27	0.17	0.20
Countries S3	inf	1	1.00	1.00
Kinship1990_EXTENDED	0.46	23	0.13	0.34

550 C Datasets and Detailed Experiments

551 C.1 Datasets

552 We evaluate NSR on four knowledge-graph datasets of varying scale and relational complexity.
553 Nations, Kinship, and Countries S3 are standard benchmarks in the KG-embedding literature. Kin-
554 ship1990_EXTENDED is an extended derivative of the classical Kinship domain that we introduce to
555 stress-test compositional relational reasoning; its construction is detailed in § C.2. Table 4 summarizes
556 the number of entities, relation types, and triples (edges) in each split.

Table 4: Statistics of the datasets used in our experiments. All counts in the last five columns refer to the number of edges (triples).

Dataset	Entities	Relations	Total	Train	Val	Test
Nations	14	55	1,992	1,592	199	201
Kinship	104	26	10,686	8,544	1,068	1,074
Kinship1990_EXTENDED	480	14	2,240	1,568	224	448
Countries_S3	271	2	1,033	985	24	24

557 C.2 How to Construct Kinship1990_EXTENDED

558 **Original data and its limitations.** The original Kinship dataset [27] contains 112 triples over 24
559 persons from two isomorphic families, with 12 binary relations: wife, husband, mother, father,
560 daughter, son, sister, brother, aunt, uncle, niece, and nephew. Three features limit its
561 utility for testing compositional reasoning: (i) *No symmetric relations*—every relation is directional,
562 so link-prediction can be solved by learning inverses alone; (ii) *Flat relation labels*—genealogical
563 concepts such as aunt or uncle are provided as atomic symbols rather than explicit compositions of
564 primitives (e.g., mother $\circ$ sister), so models are not required to discover relational grammar; (iii)
565 *Small scale*—104 triples provide insufficient coverage to stress-test generalization.

Reconstructing the base graph. Because the two families have identical structure, the complete genealogical tree is fully determined by the original triples. We first reconstruct this tree, obtaining all parent-child and spousal links.

Injecting symmetric relations. We add `sibling` as an explicitly symmetric relation. For every pair of distinct children sharing at least one parent in the reconstructed tree, we insert `sibling(X, Y)` and `sibling(Y, X)`. Unlike the original gender-specific `brother` and `sister`, `sibling` forces the model to respect an equivalence constraint that cannot be reduced to a directional inverse.

Extracting compositional chain relations. We define five new relation types by enumerating multi-hop genealogical paths in the reconstructed tree and abstracting them into explicit labels:

- `grandmother_chain`: the mother of one’s father or mother;
- `grandfather_chain`: the father of one’s father or mother;
- `maternal_aunt_chain`: the sister of one’s mother;
- `paternal_uncle_chain`: the brother of one’s father;
- `cousin_chain`: the child of one’s parent’s sibling.

For example, `cousin_chain(h, t)` is generated whenever there exist intermediates x, y such that x is a parent of h , y is a sibling of x , and t is a child of y . These chain relations are not redundant with the original flat labels; they make compositional substructure explicit and therefore require the model to reuse primitive relations rather than memorize atomic mappings.

Statistics and splits. The extended graph contains 480 entities, 14 relation types, and 2,240 triples. We follow the standard filtered link-prediction protocol: 80% train, 10% validation, 10% test. Because the underlying family tree is small and structurally deterministic, all base-relation edges necessary for genealogical consistency appear in the training split; held-out test triples are drawn predominantly from the extended relation set so that accurate prediction benefits from compositional reuse of the trained primitives.

C.3 Detailed Experiments

Embedding-Based baseline implementations. All neural baselines are trained with PyKEEN v1.10 [2] under identical data splits. Table 5 lists the architecture and optimization settings shared across embedding models. Embedding dimension is fixed to 100. DistMult, ComplEx, RESCAL, and ConvE use the LCWA training loop; TransE and RotatE use sLCWA. ConvE requires inverse triples (`create_inverse=True`) per PyKEEN’s implementation and uses 32 output channels with dropout rates $\{0.2, 0.2, 0.3\}$. The optimizer is Adam with learning rate 10^{-3} for all models except RotatE (5×10^{-4}). Training runs for at most 150 epochs with batch size 32.

Table 5: Neural baseline hyperparameters and training configuration.

Model	Dim	Loop	Inverse	LR	Epochs	Special
TransE	100	sLCWA	No	$1e-3$	150	<code>scoring_fct_norm=1</code>
DistMult	100	LCWA	No	$1e-3$	150	—
ComplEx	100	LCWA	No	$1e-3$	150	—
RotatE	100	sLCWA	No	$5e-4$	150	—
ConvE	100	LCWA	Yes	$1e-3$	150	<code>out_ch=32, dropouts {0.2, 0.2, 0.3}</code>
RESCAL	100	LCWA	No	$1e-3$	150	—

Rule Learning baseline implementations. RNNLogic operates *without pretrained knowledge-graph embeddings*. It first mines relational paths up to length 3 using the C++ miner (16 threads), then learns rule weights via a lightweight predictor trained for 10 iterations (Adam, learning rate 5×10^{-3} , weight decay 0, hidden dimension 32, batch size 16). Label smoothing (0.1) is applied during training, and expectation-based ranking (`expectation=True`) is used at evaluation. Unlike neural baselines, no entity or relation embeddings are supplied or learned.

604 **Reproducibility.** We used five random seeds: [42, 43, 44, 45, 46], and the performance of these
605 baseline models and NSR is reported in § 4.2. PyKEEN handles its own internal seeding for
606 negative sampling and parameter initialization. NSR and all baselines are evaluated on identical
607 train/validation/test splits for each dataset.

Table 6: Symbolic baseline hyperparameters and training configuration.

Model	Rule Len	Hidden Dim	LR	Iters	Special
RNNLogic (Miner)	3	—	—	—	16 threads
RNNLogic (Predictor)	—	32	5e−3	10	batch=16, smoothing=0.1, expectation=True, without embedding

608 D Other Results Demo

609 D.1 Symmetry Reasoning Trace

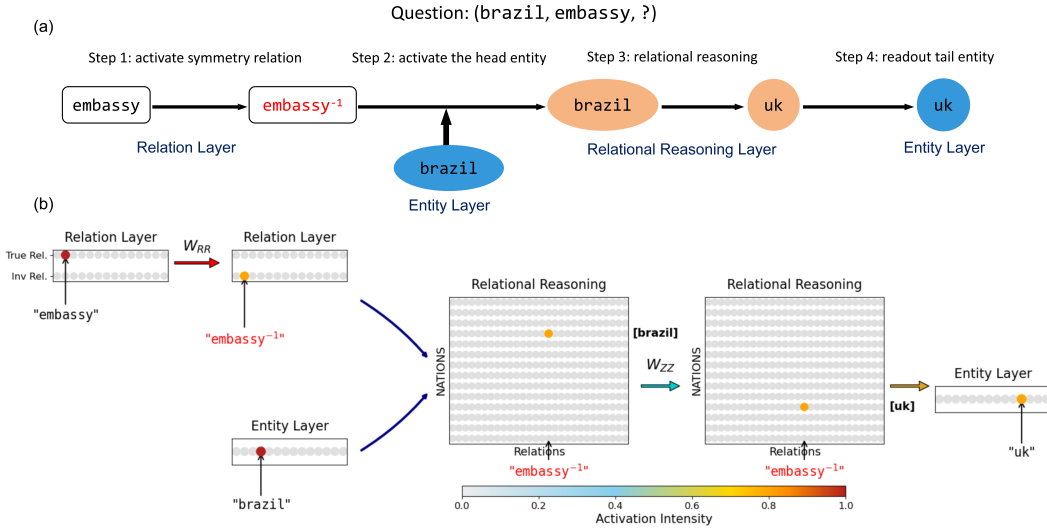


Figure 5: **Symmetry-reasoning trace (symmetry branch only).** In Path A, symmetry and inverse evaluations are performed in parallel. The complete trace therefore contains two branches: a symmetry branch that retains the original relation neuron, and an inverse branch that activates the inverse-relation neuron. For visual clarity, only the symmetry branch is shown here; the inverse branch follows the same gated-diffusion and binding structure with relation inversion.

610 Figure 5 illustrates the activation trajectory for symmetry reasoning on the query (Brazil,
611 embassy, ?). During the associative phase, the query relation embassy diffuses through W_{RR} ,
612 and the complementary gate promotes both symmetric and inverse counterparts in parallel. The full
613 inference thus comprises two structurally isomorphic branches: one retaining the original relation
614 embassy (symmetry) and one activating $embassy^{-1}$ (inverse). Because the two branches differ only
615 in whether the relation neuron is inverted, we display only the symmetry branch for clarity; the
616 inverse branch can be obtained by following the same pipeline with the W_{RR} dynamics.

617 D.2 Latent Relational Structure of Countries

618 The Countries dataset is built from only two primitive relations, `locatedIn` and `neighborOf`,
619 distributed over three entity strata: countries, regions, and continents. Crucially, `neighborOf` edges

are observed *only* between country-level entities; the dataset contains no direct adjacency between regions or between continents, even though such adjacencies are clearly entailed by the underlying geography. Recovering these missing higher-order neighborhoods therefore provides a natural probe of NSR’s compositional generalization, requiring the model to abstract the same adjacency relation across the entity hierarchy rather than within a fixed layer.

Two chained rules suffice to lift adjacency upward across strata:

$$\text{nb}(c_a, c_b) \wedge \text{loc}(c_a, R_a) \wedge \text{loc}(c_b, R_b) \Rightarrow \text{nb}(R_a, R_b), \quad (10)$$

$$\text{nb}(R_a, R_b) \wedge \text{loc}(R_a, K_a) \wedge \text{loc}(R_b, K_b) \Rightarrow \text{nb}(K_a, K_b), \quad (11)$$

where $\text{nb} \equiv \text{neighborOf}$ and $\text{loc} \equiv \text{locatedIn}$; $c_\bullet$, $R_\bullet$, and $K_\bullet$ index countries, regions, and continents respectively.

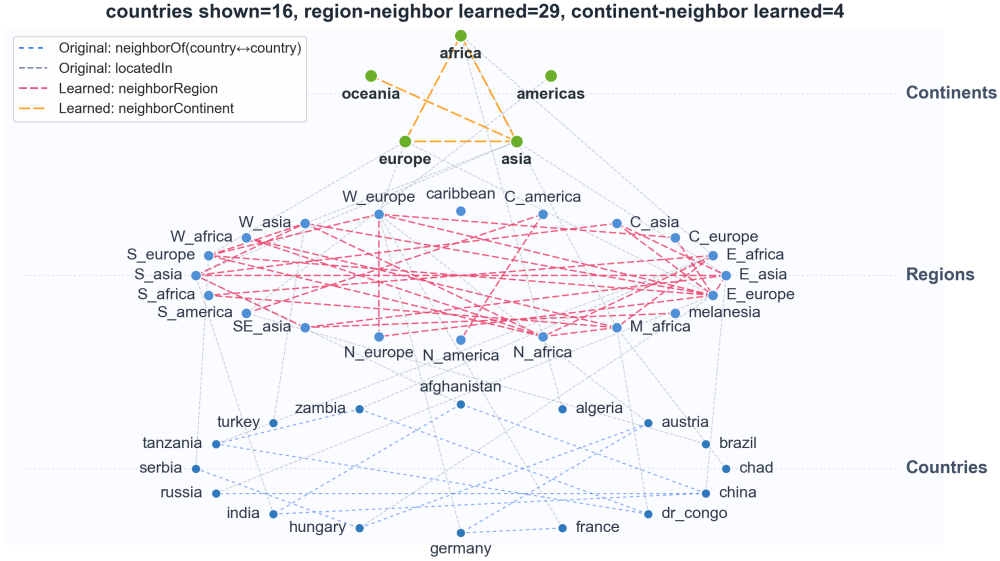


Figure 6: **Discovery of latent neighborOf on Countries.** Blue and gray dashed edges denote the original neighborOf and locatedIn facts; red and orange dashed edges denote the region- and continent-level adjacencies induced by NSR. Region prefixes abbreviate cardinal directions: N (northern), S (southern), E (eastern), W (western), C (central), M (middle), SE (southeastern). Only the 16 most active countries are shown; all regions and continents are displayed in full.

Figure 6 renders the dataset as a three-tier layered graph. Without ever observing a single direct edge of either type during training, NSR recovers 29 region-neighbor and 4 continent-neighbor edges, and the induced adjacencies are geographically coherent: the continent layer assembles the expected Africa–Europe–Asia triangle together with an Asia–Oceania link, while the region layer reproduces dense intra-Europe, intra-Africa, and intra-Asia neighborhoods (e.g., western_europe adjoining northern_europe and southern_europe).

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Justification: All training and test details are specified in the paper. The datasets and the filtered link-prediction protocol are described in § 4.1, with the construction of Kinship1990_EXTENDED detailed in § C.2. NSR’s hyperparameters for each benchmark are listed in § B.2, together with the Optuna search ranges used to obtain them and a brief sensitivity analysis. Baseline architectures and optimization settings (embedding dimension, training loop, inverse-triple usage, learning rate, batch size, epochs, and model-specific options) are documented in § C.3. The early-stopping criterion (PyKEEN’s built-in early stopper with validation MRR, patience 10, relative delta 0.002), the evaluation procedure (PyKEEN’s RankBasedEvaluator with filtering), and the use of five independent random seeds (42, 43, 44, 45, 46) are also reported in § C.3. Final results are reported as mean $\pm$ standard deviation across these five runs.

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Answer: [Yes]

Justification: All quantitative results in Tables 1 and 2 are reported as mean $\pm$ standard deviation over five independent runs with random seeds {42, 43, 44, 45, 46}. For the KG embedding baselines, variability arises from random initialization and negative-sampling order; NSR is deterministic given the data, yet we include its run-to-run timing variation. Standard deviations are reported in the same units as the means (percentage points for Hits@ k , absolute scale for MRR and MR), allowing direct assessment of statistical stability. The experimental protocol—identical train/validation/test splits, early-stopping on validation MRR, and filtered ranking—is held constant across all seeds to isolate initialization effects.

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Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

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Justification: All experiments were conducted on a single workstation equipped with an Intel Core Ultra 9 285K CPU (24 cores), 192 GB of system RAM, and a single NVIDIA RTX 6000 Ada Generation GPU (97 GB VRAM) running Linux and PyTorch. For NSR, the most time-consuming step is hyperparameter tuning via Optuna on the dataset, taking 30–60 s; the peak memory usage occurs during tuning and learning on Kinship1990_EXTENDED, reaching 994.06 MB.

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